



MALLA REDDY TECHNICAL CAMPUS
A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH
Deemed to be University
Approved by UGC & AICTE New Delhi
Maisammaguda, Medchal (Dist), Hyderabad - 500100, Telangana



Computer Aided Engineering Graphics (25CAG101)

A.Y:2025-2026

I-B.Tech I-Sem II-Mid Question Bank with Answers

Branch: CSE/CSD

Module-III

Short Questions with Answers

- 1. What is the position of the XY line in first angle projection?**

Solution: In first angle projection, the XY line acts as the reference line separating front view (above XY) and top view (below XY).

- 2. What is the first angle projection method?**

Solution: In the first angle method, the object lies in the first quadrant, i.e., above HP and in front of VP. The top view is placed below the reference line (XY), and the front view is placed above the XY line

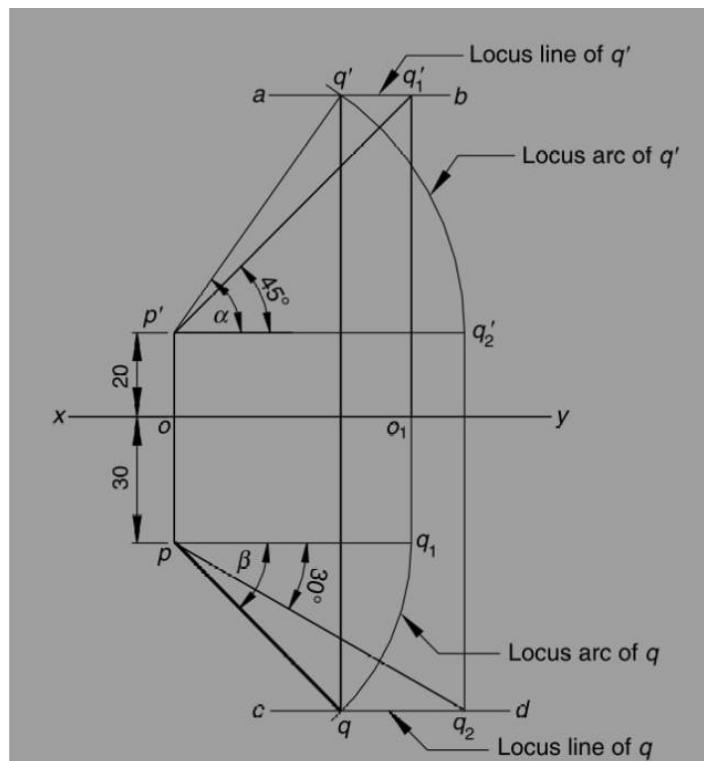
- 3. How is the true inclination with HP denoted and where is it seen?**

Solution: The true inclination with HP is denoted by θ and is seen in the front view when the line is rotated to lie parallel to VP

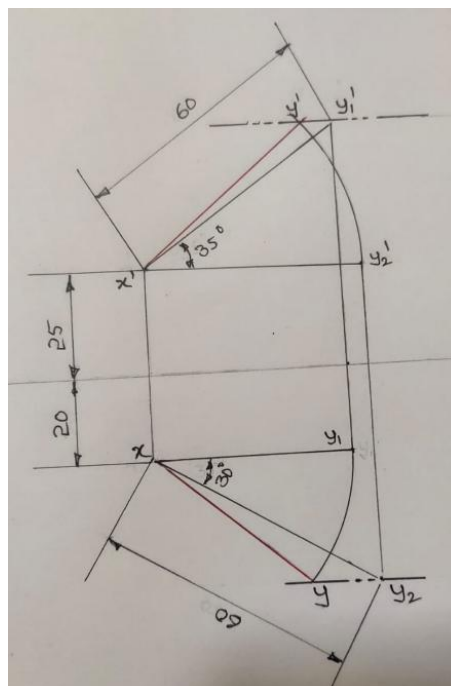
Long Questions with Answers

1. A 70 mm long line PQ has its end P 20 mm above the H.P. and 30 mm in front of the V.P. The line is inclined at 45° to the H.P. and 30° to the V.P. Draw its projections

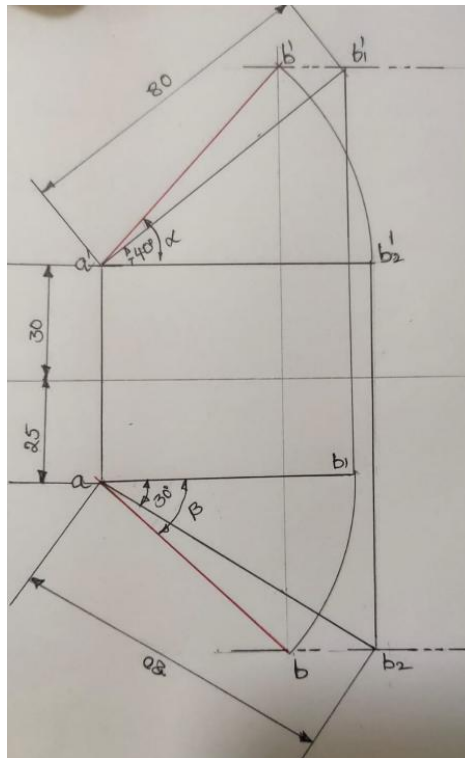
Solution



2. A 60 mm long line XY has its end X 25 mm above the H.P. and 20 mm in front of the V.P. The line is inclined at 35° to the H.P. and 30° to the V.P. draw its projections.



3. A 80 mm long line AB, has its end A 30 mm above the H.P. and 25 mm in front of the V.P. The line is inclined at 40° to the H.P. and 30° to the V.P. Draw its projections



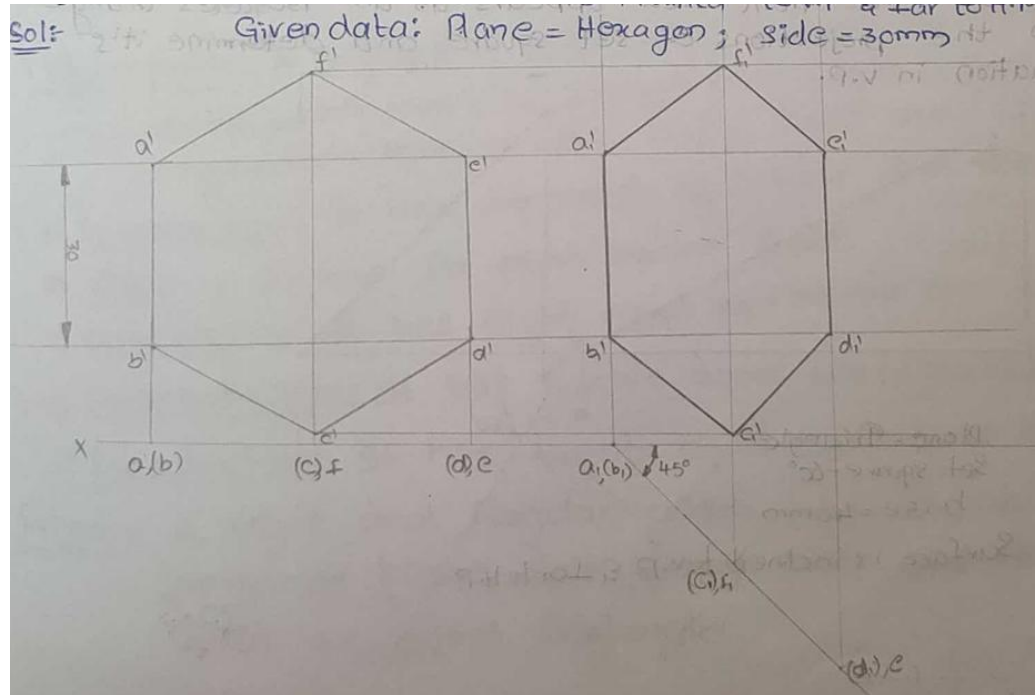
Module-IV

Short Questions with Answers

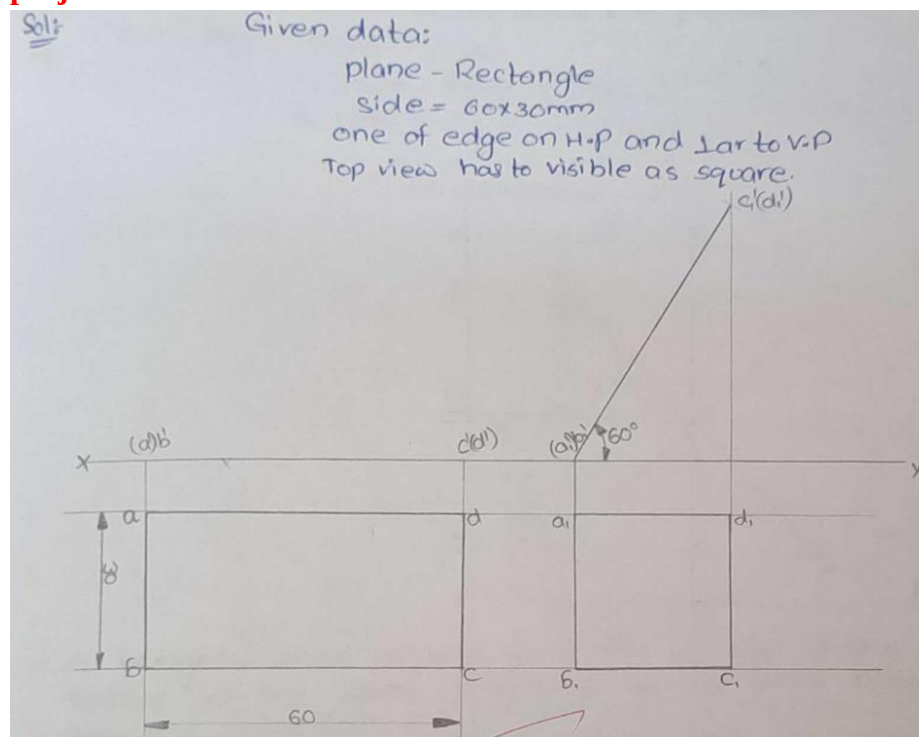
1. Define true shape of a plane.
Solution: True shape is the actual shape seen when the plane is parallel to the plane of projection
2. Which view shows the true shape of base?
Solution: The view drawn on the plane parallel to the base shows its true shape.
3. How is inclination with VP represented?
Solution: Inclination with VP is represented by ϕ and seen in the top view.
4. What is a plane in engineering graphics?
Solution: A plane is a two-dimensional flat surface having length and breadth but no thickness
5. What is a solid in engineering graphics?
Solution: A solid is a 3D object having length, breadth, and height.
6. What happens when a solid is inclined to HP?
Solution: Its front view becomes foreshortened, and the top view shows true shape of base.

Long Questions with Answers

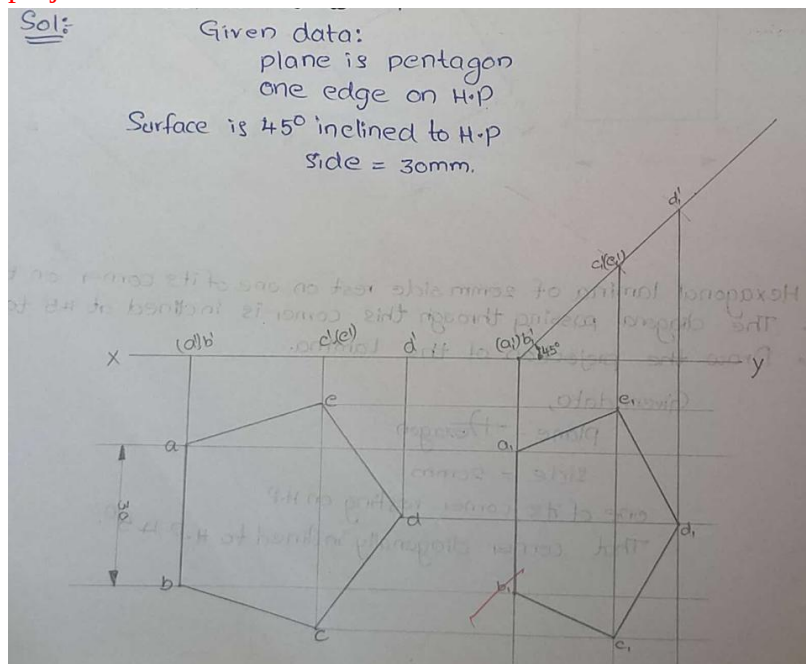
1. A hexagonal plane of side 30mm has an edge in the VP. The surface of the plane is inclined at 45° to VP and perpendicular to HP. Draw the projections.



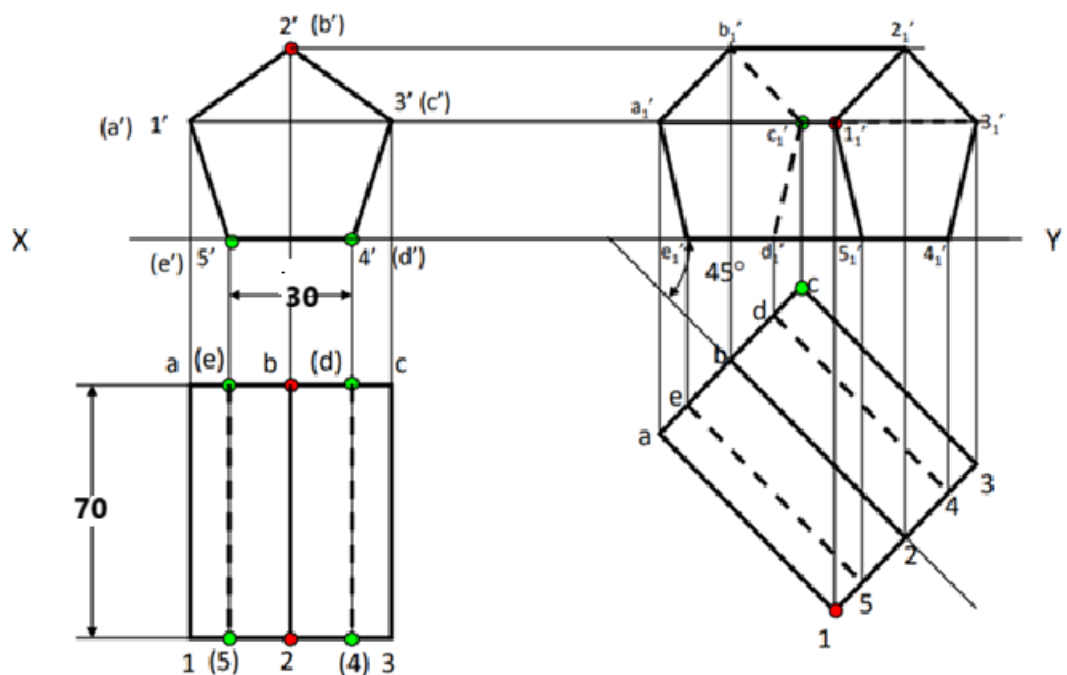
2. A rectangular lamina of side 60mmX30mm is seen as a square in top view, when it rest on one of it edge on HP and perpendicular to VP draw the projection of the lamina.



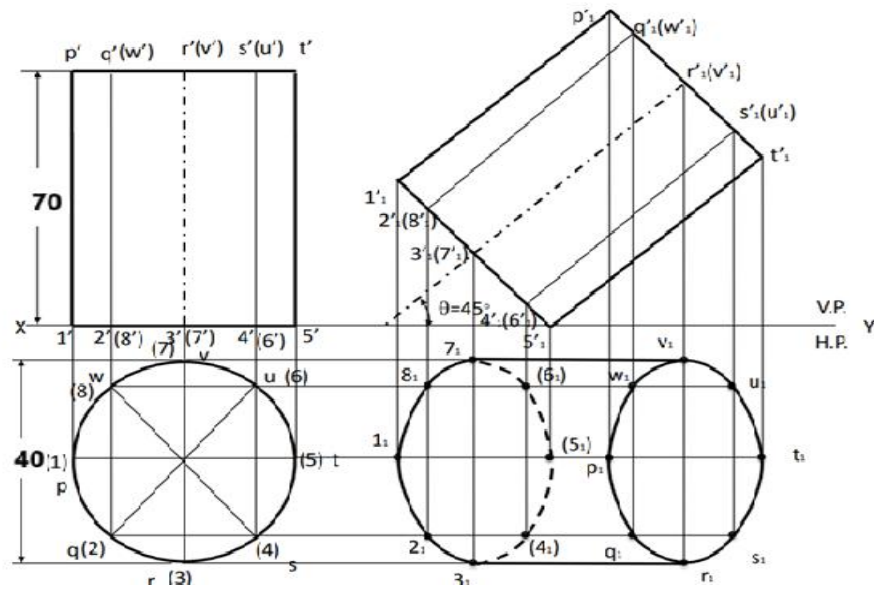
3. A regular pentagonal lamina of 30 mm side has one edge in HP draw its projections when its surface is inclined at 45° to HP.



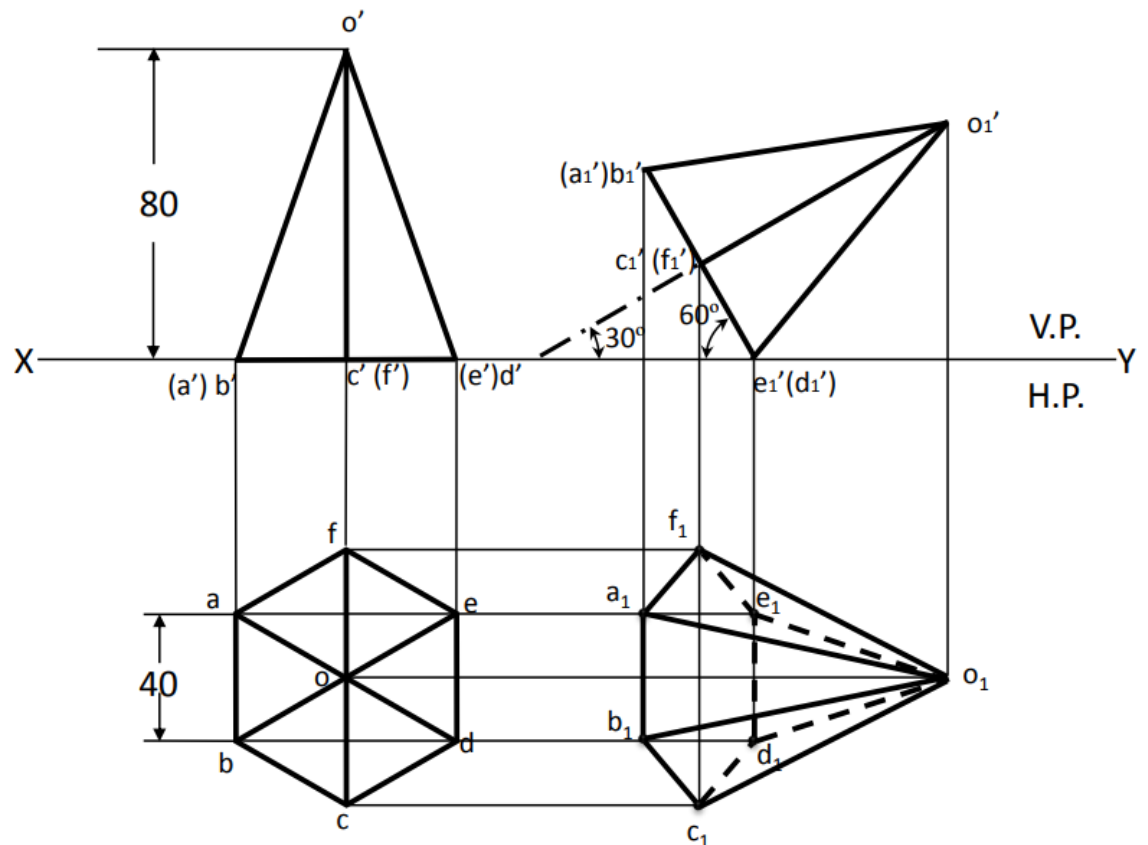
4. Draw the projections of a pentagonal prism, base 30 mm side and axis 70 mm long, resting on one of its rectangular faces on the H.P. with the axis inclined at 45° to the V.P.



5. A cylinder of diameter 40 mm and axis length 70 mm is resting on the HP on a point so that its axis is inclined at 45° to the HP and parallel to the VP. Draw its top and front views.



6. A hexagonal pyramid of base edge 40 mm and altitude 80 mm rests on one of its base edges on the HP with its axis inclined at 30° to the HP and parallel to the V.P. Draw its top and front views.



Module- V
Short Questions with Answers

1. Define orthographic view.

Solution: Orthographic view is a 2D representation of a 3D object using multiple views such as front, top, and side.

2. What is meant by converting orthographic views to isometric view?

Solution: It is the process of combining front, top, and side views to create a 3D pictorial representation of the object

3. What is an isometric axis?

Solution: Isometric axes are the three lines drawn 120° apart representing the width, depth, and height of the object

4. What is an isometric view?

Solution: An isometric view is a pictorial drawing where all three axes (X, Y, Z) are inclined at 120° to each other, giving a 3D appearance.

5. Where is the front view placed in orthographic projection?

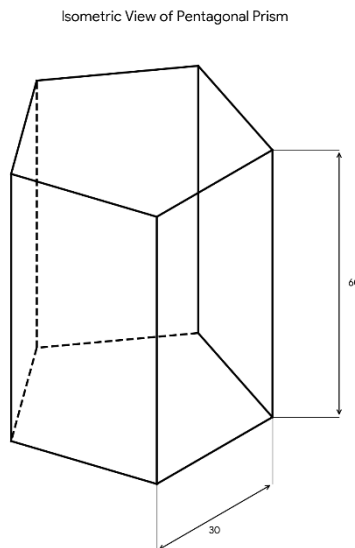
Solution: In first-angle projection, the front view is placed at the center, with the top view below it.

6. What angle do isometric axes make with each other?

Long Questions with Answers

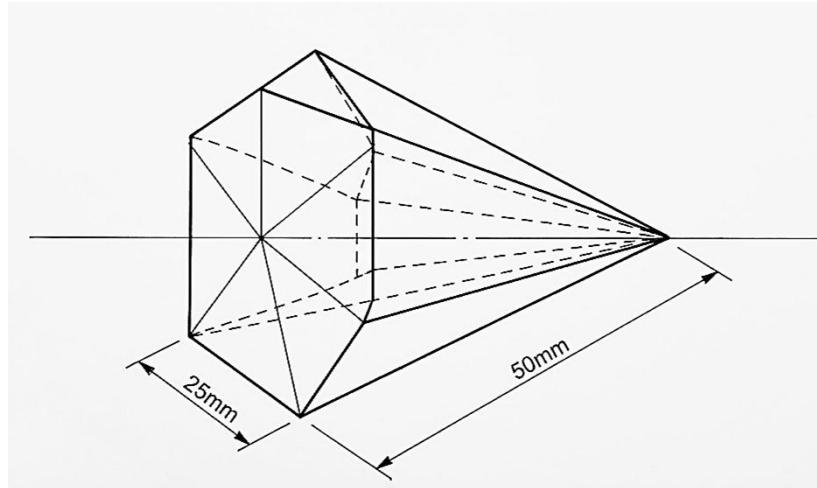
1. Draw the isometric view of a pentagonal prism, with side of base 30 mm and length of axis 60 mm, when its axis is vertical.

Solution:

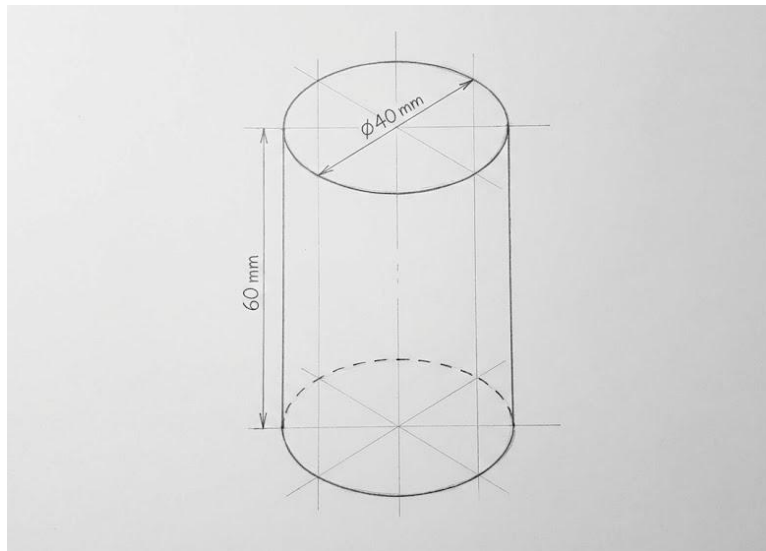


2. A hexagonal pyramid of side 25 mm, axis length 50 mm is placed on HP. Draw the isometric projection of solid when its axis is horizontal.

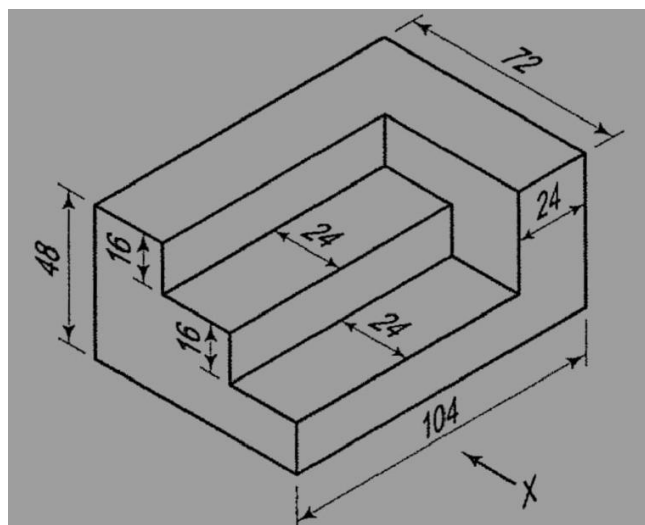
Solution:



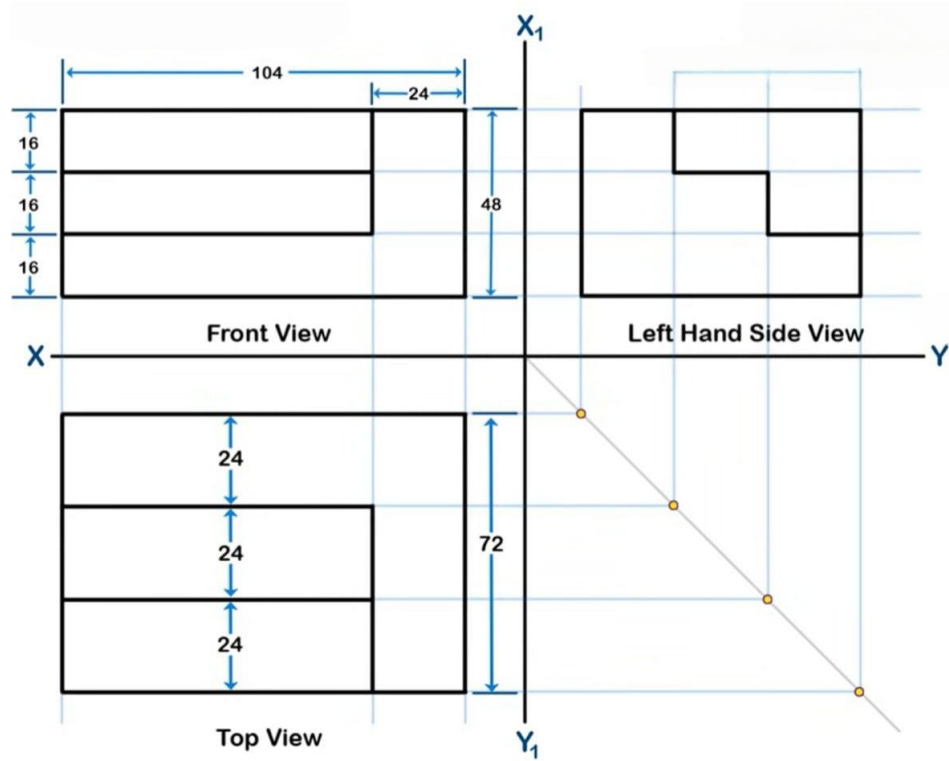
3. Draw the isometric view of cylinder of diameter 40 mm and axis 60 mm long.
Solution:



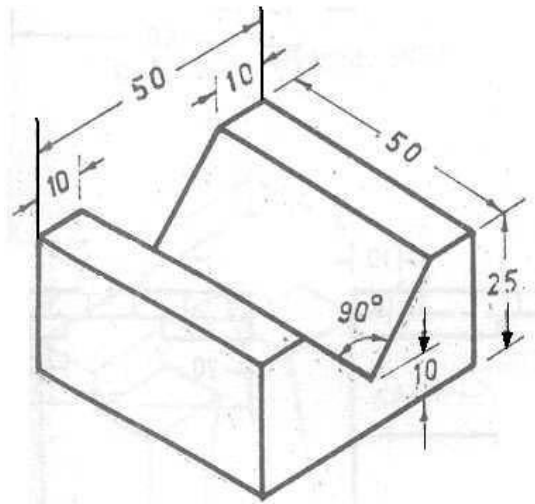
4. Draw Front view, Top view and Side view of the given isometric view given in figure



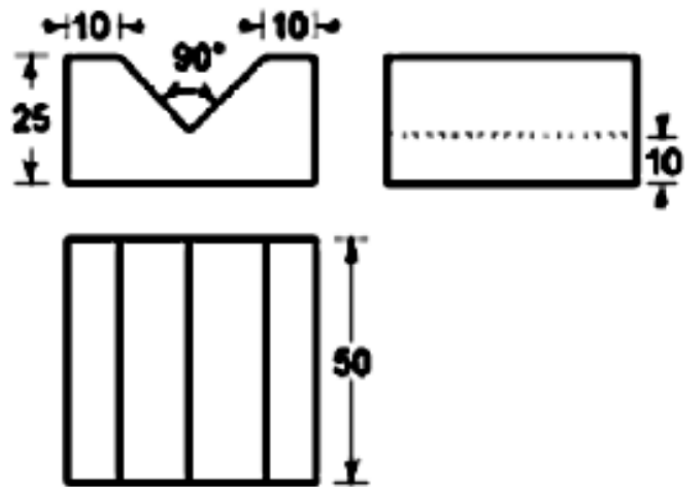
Solution:



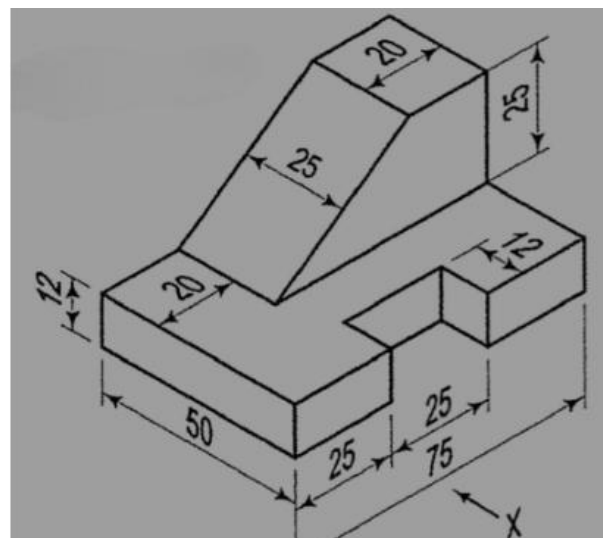
5. Draw the front view, top view and left side views of V- block as shown in figure



Solution:



6. Draw Front view, Top view and Side view of the given isometric view given in figure.



Solution:

